

Lightweight Attentive Residual U-Net (LAR-U-Net) for Image Denoising

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Abstract—Digital image acquisition and transmission are frequently compromised by stochastic variations in pixel intensity, collectively known as noise [1]. This paper introduces the Lightweight Attentive Residual U-Net (LAR-U-Net) model, an advanced image restoration architecture designed for grayscale Gaussian denoising [2]. The proposed network integrates a symmetric encoder-decoder U-Net backbone with skip connections, a Convolutional Block Attention Module (CBAM) embedded within each encoder layer, and a residual learning strategy optimized via Mean Squared Error (MSE) loss [2, 3]. To prevent the over-smoothing of critical structural features, the architecture utilizes Gradient-Augmented Processing by incorporating Sobel filter magnitude maps directly into the input stage [5]. Evaluated on the Berkeley Segmentation Dataset 500 (BSDS500), the model effectively generalizes to additive white Gaussian noise ($\sigma = 25$), achieving a minimal validation loss of 0.00327 in MSE and a Mean Absolute Error (MAE) of 0.04253 on test data [2, 6]. LAR-U-Net achieves an average Peak Signal-to-Noise Ratio (PSNR) of 19.921 dB across the entire test suite, with localized reconstruction improvements exceeding +7 dB on individual image samples. Robustness testing against the Multi Noises dataset indicates highly focused performance toward Gaussian distributions due to the mutual benefits of residual mapping and batch normalization architectures [2].

Keywords—image denoising, LAR-U-Net, residual learning, convolutional block attention module, gradient augmentation, Gaussian noise.

I. INTRODUCTION

Image noise is caused by factors such as randomness and external disturbances in the imaging process, and it may lead to problems such as blurred image details, blurred edges, and color variations. Digital image acquisition and transmission are frequently compromised by stochastic variations in pixel intensity, collectively known as noise. Over the years, researchers have proposed many methods for image noise reduction [1]. Common variants include additive white Gaussian noise and Salt-and-Pepper noise. The core challenge in image processing is the 'Denoising': effectively suppressing high-frequency noise components while preserving critical structural features, such as sharp edges and fine textures. Standard linear filters often over-smooth these features, leading to a loss of information that can be detrimental to downstream AI tasks like medical image analysis or autonomous vehicle perception.

II. PROJECT OBJECTS

The goal of image denoising is to recover a clean image x from a noisy observation y which follows an image degradation model $y = x + v$. One common assumption is that v is additive white Gaussian noise (AWGN) with standard deviation σ [3]. Preserving the texture and features of an image without getting blurred after denoising is one of the primary concerns of image denoising, which drives the investigation into balanced solutions that keep the preserved features with minimized blurring.

$$y = x + v \quad (1)$$

III. LITERATURE REVIEW

A. Deep Neural Networks for Image Denoising

The evolution of digital image denoising has shifted from traditional spatial-domain filters to advanced deep learning architectures. As outlined in a comprehensive taxonomy [1], traditional filters like the Bilateral filter are computationally efficient but often result in over-smoothing and blurred edges. To overcome these limitations, discriminative models such as DnCNN [2] were introduced. DnCNN utilizes a residual learning strategy to implicitly remove noise in hidden layers, achieving high performance in Gaussian denoising with fast convergence. However, it often struggles with texture restoration and complex, spatially variant noise.

To enhance the ability of residual networks to capture spatial structure, the CBAM-DnCNN model [3] integrates a Convolutional Block Attention Module. This approach adaptively weighs feature importance across channel and spatial dimensions, making it highly effective for precision-heavy fields like medical diagnosis and automatic driving. Another advancement is the Dual Residual Attention Network (DRANet) [4], which expands network width through parallel branches. DRANet utilizes Residual Attention Blocks (RAB) to capture complementary features, addressing the "blindness" of standard CNNs toward global dependencies. While these attention-driven models offer superior feature extraction, they often involve higher computational overhead compared to plain convolutional baselines.

B. U-Net and Hybrid Architectures

The U-Net architecture remains a cornerstone for denoising due to its encoder-decoder structure and skip connections that preserve low-level spatial information. Technical analyses into the influence of hyperparameters [6] have shown that while increasing layer size improves model fit, kernel sizes must be carefully tuned to minimize the Mean Absolute Error (MAE) and prevent overfitting.

Hybrid models like BIRUNet [5] seek to improve the traditional U-Net by integrating handcrafted image priors. By feeding bilateral filter noise residue and gradient information directly into the network, BIRUNet achieves better edge preservation in high-noise scenarios. This "guided" learning approach is particularly useful for satellite imagery and medical imaging, where maintaining structural integrity is paramount. However, the performance of such models is inherently tied to the quality of the manually derived priors.

IV. MODEL DEVELOPMENT: LIGHTWEIGHT ATTENTIVE RESIDUAL U-NET (LAR U-NET)

The LAR-U-Net is an advanced image restoration architecture that combines the spatial precision of U-Nets, the feature refinement of Attention Modules, and residual learning strategies.

1) The Input: Gradient-Augmented Processing

Instead of feeding only the raw noisy image, the model utilizes Gradient-Augmented Processing. Image gradients are computed to highlight structural edges, which are then concatenated with the original image.

- Operation: Using Sobel Filters to calculate horizontal (Gx) and vertical (Gy) gradients.

- Math: The gradient magnitude G is given by $G = \sqrt{Gx^2 + Gy^2}$.

$$G = \sqrt{Gx^2 + Gy^2} \quad (2)$$

- Why? This forces the model to recognize "edges" immediately, preventing the over-smoothing common in basic CNN-based denoisers.

2) The Backbone: U-Net with Skip Connections

The architecture adopts the Symmetric Encoder-Decoder design of a U-Net to maintain high-resolution spatial data.

- Kernel Choice: 3×3 kernels are used, balancing error reduction and model fit [6].
- Skip Connections: These bridge the encoder and decoder, ensuring that fine details lost during compression are recovered during upsampling.
- Residual Units: Constructed with residual blocks to facilitate information propagation.

3) Simple Attention Blocks (CBAM)

To prioritize important features, a Convolutional Block Attention Module (CBAM) is inserted into each encoder layer [3].

- Channel Attention: Identifies feature maps containing high-value signal versus noise. Utilizes AdaptiveAVGPool2d reduction followed by fully

connected weight mappings scaled under a Sigmoid activation boundary [3].

- Spatial Attention: Maps structural noise vulnerability zones. Compares average and maximum channel properties processed via a 7×7 convolution frame mapped down to an active Sigmoid layout [3].

4) The Output: Residual Learning Strategy

Following the DnCNN [2] paradigm, the architecture maps structural noise patterns $R(y)$ rather than direct clean pixels.

- Mechanism: Subtracts predicted noise maps from raw distorted inputs:

$$\hat{x} = y - R(y) \quad (3)$$

- Advantage: Guarantees significantly accelerated optimization properties and better runtime stability.

V. DATA ACQUISITION AND PREPROCESSING

Images are sourced from Kaggle using the Berkeley Segmentation Dataset 500 (BSDS500) [7], consisting of 500 natural images split into disjoint training, validation, and testing subsets. Cross-validation and robustness are expanded using the Multi Noises for Image Denoising dataset [8], which supplies high-quality images across 7 noise formulations: Gaussian, Speckle, Poisson, Multiplicative, JPEG Compression, Quantization, and Salt & Pepper. Preprocessing incorporates pixel normalization scaled to [0,1] alongside synthetic Gaussian generation models.

VI. EVALUATION AND ANALYSIS

Experiments were conducted utilizing $\sigma = 25$ additive white Gaussian noise (AWGN) on the BSDS500 dataset [7] following the baseline paradigm [2]. Training parameters consisted of 10 Epochs, a Batch size of 8, the Adam optimizer, and Mean Squared Error (MSE) loss configured to evaluate target noise maps:

$$L = \|y_i - LAR_U_Net(y_i) - x_i\|^2 \quad (4)$$

TABLE I. TRAINING LOSS METRICS

Ep.	Train (MSE)	Val (MSE)
1	0.22759	0.24682
2	0.00983	0.00924
5	0.00415	0.00414
10	0.00315	0.00327

The model achieved a minimum validation loss of 0.00327 in MSE, generating a final test Mean Absolute Error (MAE) of 0.04253. Reconstruction assessment via Peak Signal-to-Noise Ratio (PSNR) yielded an average test score of 19.9205 dB, showing localized improvement transitions exceeding +7 dB.

TABLE II. RECONSTRUCTION METRICS

Metric	S1	S2
PSNR Pre	20.21	20.19
PSNR Post	28.01	27.36
Δ PSNR	+7.79	+7.18

MAE Post

0.0306

0.0331

Extended benchmarking using multi-noise environments [8] produced positive metric shifts across Salt & Pepper and Multiplicative forms (gaining over 4 dB in PSNR bounds). However, performance advantages remain heavily tied to Gaussian targets. Because residual formulations and batch normalization align directly with Gaussian distributions, these components mutually benefit each other to isolate additive white Gaussian signals effectively [2].

VII. CONCLUSION

In this paper, we introduced the LAR-U-Net model which uses a residual U-Net framework combined with a Convolutional Block Attention Module (CBAM) [3] inside each encoder layer, optimized via MSE Loss for grayscale Gaussian image denoising. We achieved an average PSNR of 19.921 dB on the test dataset, with localized improvements exceeding +7 dB on specific samples. This lightweight, highly focused model offers low overall error structures, opening avenues for future development into multi-distribution architectural modifications.

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